



**Cambridge Instruments  
& Engg. Co.**

EST. 1963 · HOWRAH, INDIA

INSULATION TESTER · HAND-DRIVEN, BATTERY-FREE

# CIE/444

Hand-cranked analog insulation tester in anti-magnetic aluminium body — no batteries required. Available in single-range configurations from 100V to 5000V.

BATTERY-FREE

## Insulation Tester — Metal Body, Hand-Driven

A hand-driven generator-type insulation tester for direct-reading insulation resistance on machines, switchgear, cables and transformers — power is generated entirely by the operator, no batteries or external supply required.

- **No batteries required** — hand-cranked generator
- **Centrifugal governor** — holds output voltage constant
- **Anti-magnetic aluminium body** — die-cast, rugged
- **Cross-coil indicator** — anti-parallax mirror scale
- **3-second response** — quick direct reading

Supplied as base instrument only — carrying case, test leads, and range options available at extra cost.

*Manufactured in India since 1963 — trusted by various State Electricity Boards, Indian Railways, cement and metal industries, and other sectors.*



- V 100-5000V** Available ranges, specify at order
- 0 Battery-Free** Hand-cranked generator
- 3 3 Seconds** Response time

### 01 Full Specification Sheet

| PARAMETER                          | SPECIFICATION                                                           |
|------------------------------------|-------------------------------------------------------------------------|
| Body                               | Anti-magnetic aluminium                                                 |
| Generator Type                     | Hand-driven generator, centrifugal governor                             |
| Scale Length                       | 90 mm (approx.)                                                         |
| Accuracy                           | Conforms to IS: 2992-1987                                               |
| Rotation Speed                     | 160 r.p.m. or more                                                      |
| High Voltage Test                  | Min. 2000V or (rated V + 1000)V A.C. r.m.s., 1 min                      |
| Insulation Resistance Test         | Better than 50 M $\Omega$ at 500V D.C.                                  |
| Guard Terminal & Infinity Adjuster | Provided on instruments with insulation range 1000 M $\Omega$ and above |
| Response Time                      | 3 seconds                                                               |
| Range Configuration                | Multiple range options available — specify at order                     |
| Dimensions (L×W×H)                 | 110 × 128 × 195 mm (approx.)                                            |
| Weight                             | 2.5 kg (approx.)                                                        |
| Warranty                           | 12 months against manufacturing defects                                 |

*Custom ranges and configurations can also be built against specific requirements.*

### 02 Typical Applications

- Motors, generators, and other rotating machinery
- Insulation control and switching equipment checks
- Distribution network and cable insulation testing
- Transformer, insulator, and installation checks
- Locating shorts, defects, and insulation failures

### 03 Optional Accessories

- + Carrying case
- + Test leads, 1m/2m
- + H.V. test leads, 2m
- + Dual voltage/range option

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Instruments for accuracy and  
reliability since 1963.